

**S2424CD: A RANDOMIZED CONTROLLED TRIAL OF AN
INTERVENTION CALLED “ALGORITHM-ENABLED
PATIENTS ACTIVATED IN CANCER CARE THROUGH
TEAMS” (A-PACT) TO IMPROVE GOALS OF CARE
COMMUNICATION FOR PEOPLE WITH CANCER**

Implementation Guide

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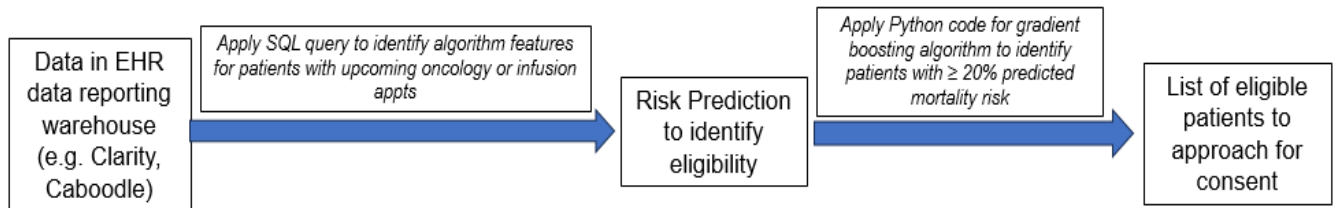
Dana-Farber Cancer Institute

Summary:

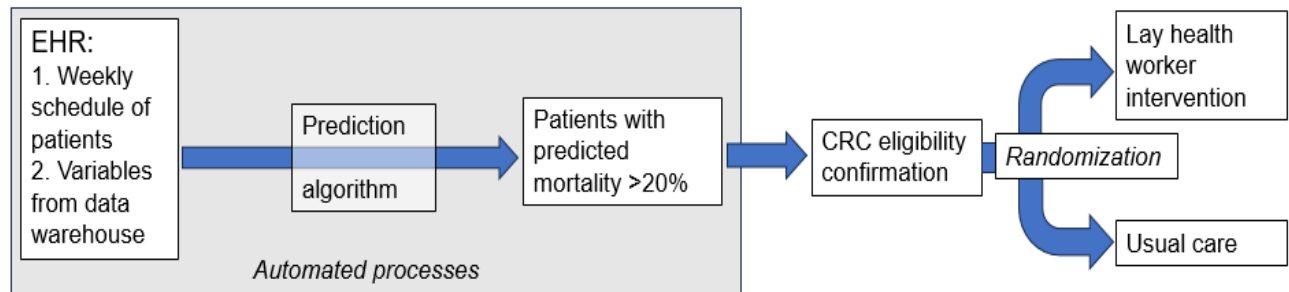
The S2424CD study uses a machine learning (ML) algorithm to identify oncology patients at high risk of short-term mortality, as a proxy for determining the need for goals-of-care (GoC) communication. The algorithm, implemented via automated queries from electronic health records (EHRs), supports weekly identification of high-risk patients. These patients are then approached for enrollment in a randomized controlled trial assessing a communication intervention. The system is designed to be locally deployed, scalable, and does not require in-EHR app development. Of particular note, no site patient-level data will need to be transferred to the study team either as part of the validation/reliability check nor as part of the study. Algorithms will be run locally by site IT teams. If feasible robustness and accuracy of the algorithm will be validated prior to and during the study.

- **Algorithm Use & Threshold:** Flags patients with $\geq 20\%$ 6-month mortality risk to identify candidates for GoC discussions; selected threshold balances sensitivity and manageability.
- **Implementation & Automation:** Weekly EHR data is extracted via standard SQL queries and processed using open-source Python code; no EHR app development required.
- **Validation & Reliability (optional):** If feasible, algorithm performance will be assessed using historical data from within the prior two years and a pre-launch “silent” run; retraining occurs if needed (AUC must exceed 0.70).
- **Patient Recruitment Process:** High-risk patients are reviewed by research staff, approached for consent, and randomized (with stratification) into control or intervention arms.
- **Ongoing Monitoring:** Performance is reviewed every 6 months to address any drift and maintain algorithm accuracy.

Algorithm Flowchart



B. Trial enrollment workflow with algorithm



SQL Query Instructions

1. Connect to the Oracle Database:

Use a database client or a tool like SQL*Plus, SQLcl, or SQL Developer to connect to the Oracle EPIC Clarity server. You'll need the necessary credentials (username, password, and database connection details).

2. Execute Your SQL Query:

Once connected, you can execute your SQL query. Make sure your query retrieves the desired features for your predictive model. For example:

```
SELECT column1, column2, ...  
FROM your_table  
WHERE your_conditions;
```

3. Export Query Results to a .csv File:

After executing the query, you can export the results to a .csv file. In SQL*Plus or SQLcl, you can use the `SPOOL` command to redirect output to a file. For example:

```
SPOOL your_output_file.csv  
SELECT column1, column2, ...  
FROM your_table  
WHERE your_conditions;  
SPOOL OFF;  
...
```

In SQL Developer, you can right-click on the result set and choose "Export Data" or use the SQL Developer Export Wizard to save the results as a .csv file.

4. Verify and Review:

Once the file is generated, open it with a text editor or spreadsheet software to verify that the data is correctly exported.

5. Additional Considerations:

- Ensure that you have the necessary permissions to run the query and write to the file.
- Be cautious about handling sensitive data and follow any data protection policies in place.

Remember to replace placeholders like `column1`, `your\_table`, and `your\_conditions` with your actual column names, table names, and conditions. If you encounter specific issues or have more detailed requirements, feel free to provide additional information for more targeted assistance.

The software to create Elixhauser comorbidities is also available at https://hcup-us.ahrq.gov/toolssoftware/comorbidityicd10/comorbidity_icd10.jsp

Elixhauser Comorbidity Software Refined for ICD-10-CM

The Elixhauser Comorbidity Indices were originally developed using ICD-9-CM diagnosis codes and adult, nonmaternal inpatient discharge data. 2 The development of the ICD-10-CM version of the indices was consistent with the methodology used for the ICD-9-CM version of the tool but used more recent 2018 data. The Elixhauser Comorbidity Indices Refined for ICD-10-CM is designed to predict two ...
hcup-us.ahrq.gov

Github

The github link below has links to our gradient boosting model. This is the algorithm that would be used to generate 6-month mortality estimates.

<https://github.com/nvelingker/eol-onc>

Next Steps

Handling High-Risks Patient Lists

Research staff conduct a weekly review of the algorithmically generated list of high-risk patients. Each identified patient undergoes clinical evaluation to confirm eligibility before any approach is made. Once eligibility is confirmed, research staff contact potential participants to explain the study and seek informed consent. Consented patients are then randomized in a 1:1 ratio to either the control group receiving usual care or the intervention group receiving the A-PACT communication intervention. The randomization process incorporates appropriate stratification factors to ensure balanced groups.

Information for Clinical Research Coordinators

Clinical research coordinators will receive weekly lists containing identified high-risk patients along with their predicted mortality risk percentages. These lists will include patient demographic information, current treatment status, contact information, and details about the treating oncologist. Coordinators will also be provided with tracking tools to monitor patient approach, consent status, and enrollment progress. Additionally, they will receive a comprehensive schedule outlining all required assessments for enrolled patients, including timeframes and procedures for follow-up.

Study Population and Assessments

The study will enroll patients with solid tumor malignancies who are either receiving or planned to receive systemic anticancer therapy and have a predicted 6-month mortality risk exceeding 20%. A total of 1000 eligible patients will be recruited, with 500 assigned to each study arm. The primary outcome measure will be the effect of the intervention on hospitalization rates. Throughout the 12-month study period, patients will complete validated questionnaires measuring mood, communication quality, and prognostic awareness at 12-week intervals.

Monitoring and Quality Control

A data safety monitoring board will conduct regular reviews of enrollment progress and outcome measures to ensure study integrity and patient safety. If enrollment is not proceeding at recommended levels, changes to the algorithm identification process and/or threshold will be considered.